

Tanaka Certificates

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Abstract

In these notes, I am building up the intuition and theory behind supermartingale certificates, with relaxed assumptions on the certificate function V . In particular, it is explored what happens when V is not twice continuously-differentiable, but instead piecewise twice continuously-differentiable.

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1 Piecewise linear certificates in dimension one

Consider the following setup:

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE
2. $V(x)$ be a piecewise linear function such that $V(x) = a_i x + b_i$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$
4. as a consequence, $V'_-(x_i) = a_{i-1}$, $V'_+(x_i) = a_i$

Now we will consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale.

Hypothesis

Does concavity of $V(x)$ (along with an interior / drift term condition) guarantee that $V(X_t)$ is a supermartingale?

To answer this, let's apply the Itô-Tanaka formula to $V(X_t)$:

$$V(X_t) = V(X_0) + \int_0^t V'(X_s) f(X_s) ds \quad (1)$$

$$+ \int_0^t V'(X_s) g(X_s) dW_s \quad (2)$$

$$+ \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X) \quad (3)$$

$$(4)$$

We will focus on these terms:

1. $D_t = \int_0^t V'(X_s) f(X_s) ds$, the drift term
2. $M_t = \int_0^t V'(X_s) g(X_s) dW_s$, the diffusion term (stochastic integral)
3. $K_t = \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X)$, the kink term

we need to prove that $D_t + M_t + K_t$ is a supermartingale.

Some natural assumptions, which are sufficient to guarantee that $V(X_t)$ is a supermartingale, are the following:

1. $f(x)V'(x) \leq 0$ for all $x \in (x_i, x_{i+1})$. This is because of the following:

Fact

If $f(X)V'(X) \leq 0$, then the drift term D_t is a supermartingale.

Proof: We want to prove $\mathbb{E}[D_t | \mathcal{F}_s] \leq D_s$ for $s \leq t$.

$$\mathbb{E}[D_t | \mathcal{F}_s] = \mathbb{E} \left[\int_0^t V'(X_u) f(X_u) du \middle| \mathcal{F}_s \right] \quad (5)$$

$$= \int_0^s V'(X_u) f(X_u) du + \mathbb{E} \left[\int_s^t V'(X_u) f(X_u) du \middle| \mathcal{F}_s \right] \quad (6)$$

$$\leq D_s + 0 = D_s \quad (7)$$

2. M_t is a martingale. Assuming $V(x)$ has finitely many pieces (as is the case for a neural network with ReLU activation functions), it is enough to impose a square-integrability condition on $g(x)$ ¹
3. $V'_+(x_i) - V'_-(x_i) \leq 0 \iff a_{i-1} \geq a_i$, i.e. the **concavity condition**. This is because of the following:

¹For example, $\mathbb{E} \left[\int_0^t g(X_s)^2 ds \right] < \infty$.

Fact

If $V'_+(x_i) - V'_-(x_i) \leq 0$ and $L_t^{x_i}(X) \geq 0$, then the kink term K_t is a supermartingale.

Proof: We want to prove $\mathbb{E}[K_t|\mathcal{F}_s] \leq K_s$ for $s \leq t$.

$$\mathbb{E}[K_t|\mathcal{F}_s] = \mathbb{E}\left[\frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X) \middle| \mathcal{F}_s\right] \quad (8)$$

$$= \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) \mathbb{E}[L_t^{x_i}(X)|\mathcal{F}_s] \quad (9)$$

$$\leq \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_s^{x_i}(X) = K_s \quad (10)$$

Note that the inequality follows from the fact that $L_t^{x_i}(X)$ is non-decreasing in t , hence $\mathbb{E}[L_t^{x_i}(X)|\mathcal{F}_s] \geq \mathbb{E}[L_s^{x_i}(X)|\mathcal{F}_s] = L_s^{x_i}(X)$.

We will consider what happens in different practical cases to these terms, including:

1. A simple one-dimensional SDE with a piecewise linear certificate.
2. A pendulum with a stochastic torque, and X_t measuring the angle of the pendulum.
3. A network b black and w white nodes, with given transition probabilities, and X_t measuring the amount of mass in the white nodes.

to ensure our assumptions are reasonable.

1.1 Proof of the supermartingale property

First, let's note the following:

Fact

The sum of a martingale and a supermartingale is a supermartingale.

Proof: By definition, given a martingale M_t and a supermartingale S_t , for $r < t$ we have $\mathbb{E}[M_t|\mathcal{F}_r] = M_r$ and $\mathbb{E}[M_t|\mathcal{F}_r] \leq S_r$. Then, for $r < t$:

$$\mathbb{E}[M_t + S_t|\mathcal{F}_r] = \mathbb{E}[M_t|\mathcal{F}_r] + \mathbb{E}[S_t|\mathcal{F}_r] \quad (11)$$

$$= M_r + \mathbb{E}[S_t|\mathcal{F}_r] \quad (12)$$

$$\leq M_r + S_r = (M + S)_r \quad (13)$$

hence $M_t + S_t$ is a supermartingale.

The drift term D_t is a supermartingale by assumption 1. The kink term K_t is a supermartingale by assumption 3. The stochastic integral M_t is a martingale by assumption 2. Therefore, the sum of these three terms is a supermartingale (as a consequence of the previous fact), and hence $V(X_t)$ is a supermartingale.

2 Piecewise $\mathcal{C}^2$ certificates in dimension one

The setup is pretty similiar, with the exception that V is now piecewise $\mathcal{C}^2$ instead of piecewise linear.

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE.
2. $V(x)$ be a piecewise $\mathcal{C}^2$ function such that $V(x) \in \mathcal{C}^2$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$.

We will again consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale. Again, the natural hypothesis is:

Hypothesis

Does concavity of $V(x)$ (along with an interior / drift term condition) guarantee that $V(X_t)$ is a supermartingale?

We apply the Itô-Tanaka formula to $V(X_t)$, and we get similar terms, with the addition of

$$C_t = \frac{1}{2} \int_0^t V''(X_s) g^2(X_s) ds \quad (14)$$

Since the integral in the C_t term is over the same intervals as the D_t term, we can combine them to recover the generator / Lie derivative term:

$$D_t + C_t = \int_0^t \left(V'(X_s) f(X_s) + \frac{1}{2} V''(X_s) g^2(X_s) \right) ds = \int_0^t \mathcal{L}V(X_s) ds \quad (15)$$

The proof that $V(X_t)$ is a supermartingale is similar to the piecewise linear case, but now we need to impose the condition:

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g^2(x) \leq 0 \quad (16)$$

Table 1: Piecewise linear certificates examples.

SDE	$V(x)$	D_t	M_t	K_t
$dX_t = dW_t$	$- x $	0	$\int_0^t -\text{sgn}(X_s) dW_s$	$-L_t^0(X)$
$dX_t = -X_t dt$	$ x $	$-\int_0^t X_s ds$	0	0

Table 2: Piecewise $\mathcal{C}^2$ certificates examples.

SDE	$V(x)$	D_t	M_t	C_t	K_t
$dX_t = -\lambda X_t dt + \sigma dW_t$	x^2	$-2\lambda \int_0^t X_s^2 ds$	$\int_0^t 2\sigma X_s dW_s$	$\sigma^2 t$	0
$dX_t = dt + \sqrt{2} dW_t$	$x - \frac{x^2}{2}$	$\int_0^t (1 - X_s) ds$	$\int_0^t (1 - X_s) \sqrt{2} dW_s$	$-t$	0
$dX_t = dW_t$	$\begin{cases} -\frac{1}{2}x^2, & x < 0, \\ -x, & x \geq 0 \end{cases}$	0	$\int_0^t V'(X_s) dW_s$	$-\frac{1}{2} \int_0^t \mathbf{1}_{\{X_s < 0\}} ds$	$-\frac{1}{2} L_t^0(X)$

3 Studying piecewise certificates on concrete one-dimensional examples

Some useful examples to study are given in Tables 1 and 2.

3.1 Ornstein–Uhlenbeck process in three parameter regimes

Consider the Ornstein–Uhlenbeck process

$$dX_t = -\lambda X_t dt + \sigma dW_t$$

with the quadratic certificate $V(x) = x^2$. Its generator is

$$\mathcal{L}V(x) = -2\lambda x^2 + \sigma^2 = -2\lambda \left(x^2 - \frac{\sigma^2}{2\lambda} \right).$$

Consequently, if

$$r = \frac{\sigma}{\sqrt{2\lambda}},$$

then the generator is negative for $|x| > r$, zero for $|x| = r$, and positive for $|x| < r$. The decomposition is

$$X_t^2 - X_0^2 = \underbrace{-2\lambda \int_0^t X_s^2 ds}_{D_t} + \underbrace{2\sigma \int_0^t X_s dW_s}_{M_t} + \underbrace{\sigma^2 t}_{C_t},$$

with $K_t = 0$.

Figure 1 fixes $\lambda = 1$ and $X_0 = 1$, and uses $r = 0.5$, $r = 1$, and $r = 1.5$. These choices make the initial generator negative, zero, and positive, respectively. Moreover,

$$\mathbb{E}[X_t^2] = r^2 + (X_0^2 - r^2)e^{-2\lambda t},$$

so the expected generator retains the corresponding sign. In the middle case, X_t^2 is constant in expectation, but it is not a martingale: the pointwise generator vanishes only when $|X_t| = r$.

The stochastic integral M_t need not visibly oscillate on every individual sample path; the martingale statement is instead that its conditional drift is zero. Figure 1 therefore plots Monte Carlo expectations. The mean of M_t remains statistically consistent with zero, while D_t and C_t determine whether $\mathbb{E}[X_t^2 - X_0^2]$ decreases, remains zero, or increases.

3.2 Failure of piecewise linearity and success of curvature

For $f = 1$ and $g = \sqrt{2}$ on $[0, 1]$, consider a case where want to find a certificate V such that $V(0) = 0$ and $V(1) = 1$.

OU process with $V(x) = x^2$, $\lambda = 1$, and $X_0 = 1$

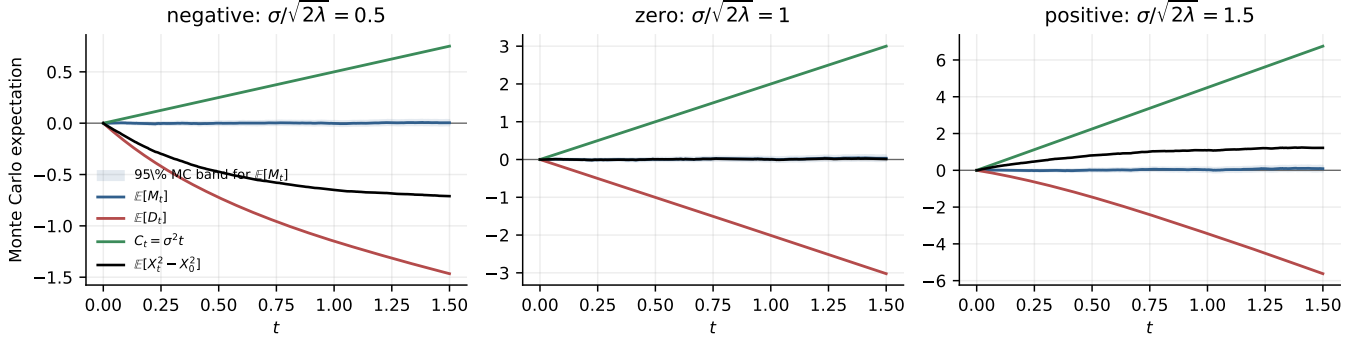


Figure 1: Monte Carlo expectations from 8000 Ornstein-Uhlenbeck paths for $V(x) = x^2$. The three columns use $\sigma/\sqrt{2\lambda} = 0.5, 1, 1.5$. The shaded region is a pointwise approximate 95% Monte Carlo band for $\mathbb{E}[M_t]$.

A smooth candidate could be:

$$V_{\mathcal{C}^2}(x) = x - \frac{x^2}{2},$$

the curvature contribution compensates for the positive drift:

$$\mathcal{L}V_{\mathcal{C}^2}(x) = (1 - x) + \frac{1}{2}(-1)(2) = -x \leq 0.$$

and indeed $V_{\mathcal{C}^2}$ is a valid supermartingale certificate.

On the other hand, let's assume V is piecewise linear. Then there exists an interval $(x_i, x_{i+1}) \subset [0, 1]$ where $V'(x) = a_i > 0$. On that interval, given an $x \in (x_i, x_{i+1})$, the curvature term vanishes, and the generator is

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g^2(x) = V'(x)f(x) = V'(x) = a_i > 0.$$

This example motivates the following hypothesis, which we will study in more detail in [Section 4](#).

Hypothesis

While piecewise linear certificates V_{PL} are not sufficient to guarantee that V_{PL} is a supermartingale, **piecewise quadratic certificates** are sufficient, as the curvature term can compensate for positive drift.

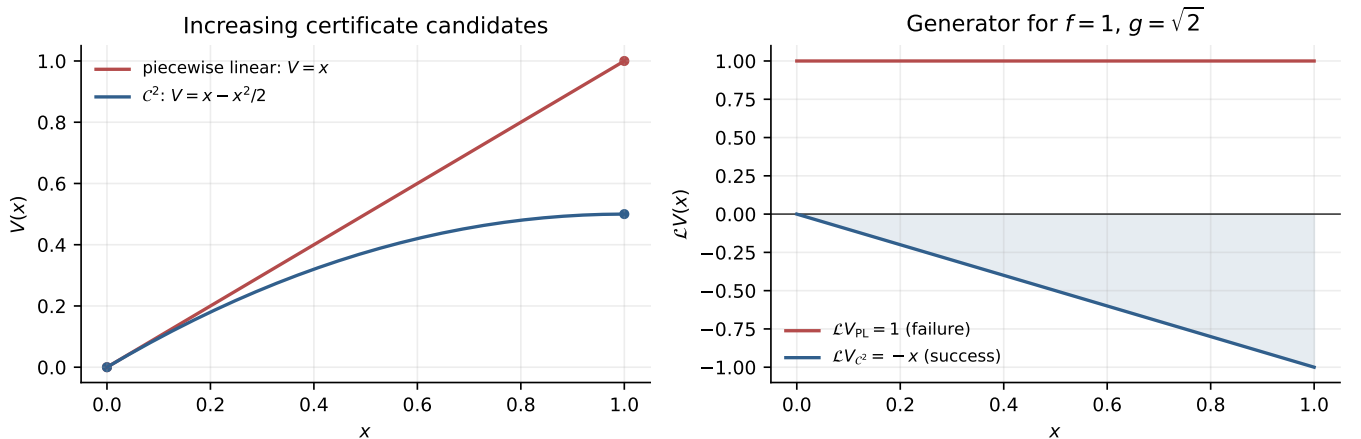


Figure 2: An increasing piecewise-linear candidate fails the generator condition for $f = 1$, $g = \sqrt{2}$, whereas negative curvature makes the smooth candidate satisfy it throughout $[0, 1]$.

3.3 A piecewise $\mathcal{C}^2$ certificate with curvature and a kink

Finally, let $dX_t = dW_t$ and

$$V(x) = \begin{cases} -\frac{1}{2}x^2, & x < 0, \\ -x, & x \geq 0. \end{cases}$$

Here $D_t = 0$, while

$$M_t = \int_0^t V'(X_s) dW_s, \quad C_t = -\frac{1}{2} \int_0^t \mathbf{1}_{\{X_s < 0\}} ds, \quad K_t = -\frac{1}{2} L_t^0(X).$$

Thus C_t decreases while the path lies below zero, and K_t decreases only when local time accumulates at the kink. Figure 3 shows these different mechanisms on the same Brownian realization. The plotted local time uses an occupation-density approximation with bandwidth 0.06.

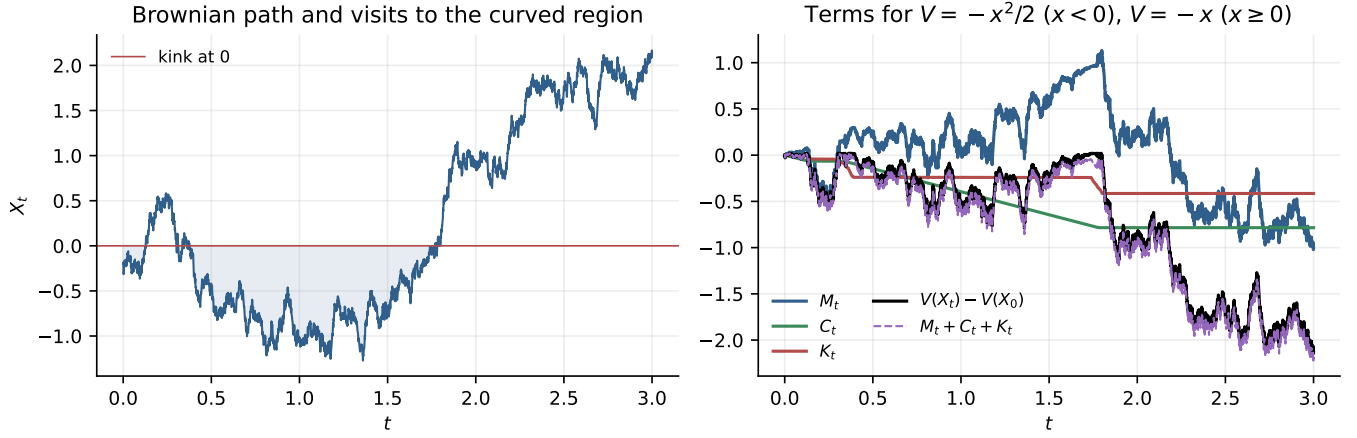


Figure 3: A Brownian realization and the terms for the hybrid piecewise $\mathcal{C}^2$ certificate. The curvature and kink terms are both nonincreasing, while the martingale term fluctuates.

4 Piecewise quadratic certificates in dimension one

In this section, we will study a failure regime of the piecewise linear certificates in dimension one and see how it leads to sufficiency of piecewise quadratic certificates.

Additionally, we observe an interesting connection to the *Taylor expansion* of the best possible certificate, namely a martingale certificate for which the generator is zero.

4.1 Failure of piecewise linear certificates in dimension one

We will consider the setup from [Section 1](#) again, but with additional assumptions on absorption at the boundary, an increase condition on the piecewise linear function $V(x)$, and a non-degeneracy condition on the diffusion term $g(x)$.

The key point is that the piecewise linear setup is not flexible enough when the certificate is required to increase in the same direction as the drift. In that case, a piecewise linear function has no curvature term which could compensate for the positive drift contribution.

So essentially, we will show that:

Fact

(*Informally.*) There is a case where no piecewise linear certificate can make $V(X_t)$ a supermartingale.

Although it should be clear from previous sections, for completeness, the setup is as follows:

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE.
2. $V(x)$ be a piecewise linear function such that $V(x) = a_i x + b_i$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$.
4. as a consequence, $V'_-(x_i) = a_{i-1}$, $V'_+(x_i) = a_i$
5. The X_t process is **absorbed** at the boundary of $I = [a, b]$.
6. $V(x)$ has the **increase condition** at the boundary:

$$V(a) < V(b). \tag{17}$$

7. $g(x)^2 > 0$ for all $x \in (a, b)$, i.e. the **diffusion term is non-degenerate**.
8. M_t is a martingale, i.e. the **martingale condition**.

Motivated by the example given in [Section 3.2](#), we will now show that there is a broad class of SDEs for which increasing piecewise linear certificates cannot be supermartingales.

Fact

Assume that for all $\varepsilon > 0$, there exists $x \in (a, a + \varepsilon)$ such that

$$f(x) > 0. \quad (18)$$

Then there is no continuous piecewise linear V satisfying the increase condition

$$V(a) < V(b) \quad (19)$$

and the concavity condition

$$V'_+(x_i) - V'_-(x_i) \leq 0 \quad (20)$$

at every breakpoint, such that $V(X_t)$ is a supermartingale.

Proof. Assume, for contradiction, that such a piecewise linear function V exists and that $V(X_t)$ is a supermartingale.

Let $p(x) = V'(x)$ on each linear region. Since V is piecewise linear, the Itô–Tanaka formula gives

$$V(X_t) = V(X_0) + \int_0^t p(X_s) f(X_s) ds \quad (21)$$

$$+ \int_0^t p(X_s) g(X_s) dW_s \quad (22)$$

$$+ \frac{1}{2} \sum_i (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X). \quad (23)$$

The stochastic integral is a martingale by assumption. Therefore, if $V(X_t)$ is a supermartingale, the finite-variation part cannot create positive drift on any linear region. Hence the necessary interior condition is

$$p(x)f(x) \leq 0 \quad \text{on every linear region.} \quad (24)$$

This also follows from [Fact 1](#) in [Section 1](#).

We now use the concavity condition. Since

$$V'_+(x_i) - V'_-(x_i) \leq 0 \quad (25)$$

at every breakpoint, the slopes of V are nonincreasing from left to right. Therefore V is concave.

Let (a, x_1) be the first linear region. We claim that the slope of V on this first region must be positive. Indeed, suppose instead that the first slope were nonpositive. Since the slopes are nonincreasing, all later slopes would also be nonpositive. Thus,

$$V(b) - V(a) = \int_a^b V'(x) dx \quad (26)$$

$$\leq 0. \quad (27)$$

This contradicts the increase condition $V(a) < V(b)$. Hence the first slope must be positive, so

$$p(x) > 0 \quad \text{for all } x \in (a, x_1). \quad (28)$$

Now use the assumption on the drift. Since for every $\varepsilon > 0$ there exists $x \in (a, a + \varepsilon)$ such that $f(x) > 0$, we may choose $\varepsilon = x_1 - a$. Then there exists $x^* \in (a, x_1)$ such that

$$f(x^*) > 0. \quad (29)$$

Combining this with (28), we get

$$p(x^*)f(x^*) > 0. \quad (30)$$

This contradicts the necessary interior condition (24). Therefore no such piecewise linear V can make $V(X_t)$ a supermartingale.

The conclusion is that piecewise linear certificates fail when the certificate is forced to increase in the same

direction as the drift. The obstruction comes from the fact that piecewise linear functions have no interior curvature term. Indeed, away from the kinks,

$$\mathcal{L}V(x) = V'(x)f(x), \quad (31)$$

so a positive drift contribution cannot be compensated. This motivates the piecewise quadratic case, where the additional curvature term

$$\frac{1}{2}V''(x)g(x)^2 \quad (32)$$

can offset the positive drift contribution.

4.2 Sufficiency of piecewise quadratic certificates in dimension one

In this subsection we will show that piecewise quadratic certificates are sufficient to guarantee that $V(X_t)$ is a supermartingale.

Let V be piecewise quadratic on $I = [a, b]$, so that on every interval (x_i, x_{i+1}) we have

$$V(x) = a_i x^2 + b_i x + c_i. \quad (33)$$

Assume also that V is continuous at the breakpoints. Then

$$V'(x) = 2a_i x + b_i, \quad V''(x) = 2a_i, \quad x \in (x_i, x_{i+1}). \quad (34)$$

Therefore the generator on the interior of a quadratic piece is

$$\mathcal{L}V(x) = (2a_i x + b_i) f(x) + a_i g(x)^2. \quad (35)$$

Here, there always exists a solution (a_i, b_i) to the interior condition $\mathcal{L}V(x) \leq 0$ on each piece. This gives the following sufficient condition.

Fact

Let V be continuous and piecewise quadratic on $I = [a, b]$. Suppose that:

1. on every open piece (x_i, x_{i+1}) ,

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g(x)^2 \leq 0; \quad (36)$$

2. at every breakpoint,

$$V'_+(x_i) - V'_-(x_i) \leq 0; \quad (37)$$

3. the stochastic integral

$$M_t = \int_0^t V'(X_s)g(X_s) dW_s \quad (38)$$

is a martingale.

Then $V(X_t)$ is a supermartingale.

Proof. By the Itô–Tanaka formula for a piecewise C^2 function,

$$V(X_t) = V(X_0) + \int_0^t \left(V'(X_s)f(X_s) + \frac{1}{2}V''(X_s)g(X_s)^2 \right) ds \quad (39)$$

$$+ \int_0^t V'(X_s)g(X_s) dW_s \quad (40)$$

$$+ \frac{1}{2} \sum_i (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X). \quad (41)$$

The first term is nonincreasing by (36). The stochastic integral is a martingale by assumption. Finally, since local time is nondecreasing in t , the kink term is nonincreasing by (37). Therefore $V(X_t)$ is a supermartingale.

4.3 Connection to Taylor expansion of the best possible certificate

Let us now again consider the SDE

$$dX_t = dt + \sqrt{2} dW_t, \quad (42)$$

the generator is

$$\mathcal{L}V(x) = V'(x) + V''(x). \quad (43)$$

A martingale certificate is obtained by asking for the generator to vanish:

$$V'(x) + V''(x) = 0. \quad (44)$$

Solving this ODE gives

$$V(x) = C_1 + C_2 e^{-x}. \quad (45)$$

If we impose the normalization $V(0) = 0$ and $V(1) = 1$, then

$$V_\star(x) = \frac{1 - e^{-x}}{1 - e^{-1}}. \quad (46)$$

This is the martingale certificate: $\mathcal{L}V_\star(x) = 0$ on the interior.

Now expand the numerator of (46) around 0:

$$1 - e^{-x} = x - \frac{x^2}{2} + \frac{x^3}{6} - \dots. \quad (47)$$

The quadratic certificate used above is exactly the second-order truncation of this numerator:

$$V_{\text{quad}}(x) = x - \frac{x^2}{2}. \quad (48)$$

So, up to the constant $(1 - e^{-1})^{-1}$ and up to truncation error, the quadratic certificate is the Taylor approximation of the martingale certificate.

5 Piecewise quadratic supermartingales in multiple dimensions

Now we will finally consider a very general setup.

First, we prove that when a piecewise-quadratic function $V : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies similar concavity and smoothness conditions as in the one-dimensional case, then $V(X_t)$ is a supermartingale.

Second, we will prove that if a smooth certificate exists, then we can construct a piecewise quadratic certificate that is a supermartingale, essentially showing how quadratic certificates are in some sense sufficient, while piecewise linear certificates are not.

5.1 Proving the supermartingale property

We will show that a piecewise quadratic function gives a supermartingale if:

1. the generator is nonpositive inside each region,
2. the stochastic integral is a martingale,
3. the normal derivative jumps in the correct direction across each interface.

Setup

Let:

1. $dX_t = f(X_t)dt + \sigma(X_t)dW_t$ be a d -dimensional SDE, where $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$.
2. W_t is an m -dimensional Brownian motion.
3. $a(x) = \sigma(x)\sigma(x)^\top$.
4. $K \subseteq \mathbb{R}^d$ is a compact set partitioned into finitely many polyhedral cells:

$$K = \bigcup_{i=1}^N K_i.$$

5. $V : K \rightarrow \mathbb{R}$ is a continuous piecewise quadratic function such that

$$V(x) = V_i(x) = c_i + p_i^\top x + \frac{1}{2}x^\top Q_i x, \quad x \in K_i,$$

where $Q_i = Q_i^\top$.

6. V is continuous across interfaces:

$$V_i(x) = V_j(x), \quad x \in K_i \cap K_j.$$

Now we will consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale, at least up to the first exit time from K .

Define the stopping time

$$\tau_K = \inf\{t \geq 0 : X_t \notin K\}.$$

Hypothesis

Does a multidimensional analogue of concavity, together with an interior generator condition, guarantee that

$$V(X_{t \wedge \tau_K})$$

is a supermartingale?

To answer this, we apply the multidimensional Itô–Tanaka formula to $V(X_{t \wedge \tau_K})$.

On each cell K_i , we have

$$\nabla V_i(x) = p_i + Q_i x, \quad \nabla^2 V_i(x) = Q_i.$$

Therefore, the infinitesimal generator of V_i is

$$\mathcal{L}V_i(x) = \nabla V_i(x) \cdot f(x) + \frac{1}{2} \text{tr}(a(x) \nabla^2 V_i(x)).$$

Substituting the quadratic form gives

$$\mathcal{L}V_i(x) = (p_i + Q_i x) \cdot f(x) + \frac{1}{2} \text{tr}(a(x) Q_i).$$

For $i \neq j$, let

$$\mathcal{F}_{ij} = \{F : F \text{ is a connected } (d-1)\text{-dimensional face of } K_i \cap K_j\}.$$

Sidenote

Each cell K_i carries its own quadratic piece V_i ; the function need not be given by one quadratic expression on $K_i \cup K_j$. Two such cells can touch along more than one $(d-1)$ -dimensional interface, or facet. The set $\mathcal{F}_{ij}$ records these connected faces separately so that the normal and the Tanaka condition can be specified on each one.

For each $F \in \mathcal{F}_{ij}$, choose a unit normal $n_{i \rightarrow j, F}$ pointing from K_i into K_j . If

$$F \subseteq \{x : h_F^\top x = \gamma_F\},$$

then one may take

$$n_{i \rightarrow j, F} = \text{sign}(h_F^\top (x_j - x_i)) \frac{h_F}{\|h_F\|},$$

where $x_i \in \text{int}(K_i)$ and $x_j \in \text{int}(K_j)$.

The multidimensional Itô–Tanaka formula gives

$$V(X_{t \wedge \tau_K}) = V(X_0) + \int_0^{t \wedge \tau_K} \mathcal{L}V(X_s) ds \quad (49)$$

$$+ \int_0^{t \wedge \tau_K} \nabla V(X_s)^\top \sigma(X_s) dW_s \quad (50)$$

$$+ \frac{1}{2} \sum_{i < j} \sum_{F \in \mathcal{F}_{ij}} \int_0^{t \wedge \tau_K} (\nabla V_j(X_s) - \nabla V_i(X_s)) \cdot n_{i \rightarrow j, F} dL_s^F. \quad (51)$$

Here L_t^F denotes the surface local time accumulated by X_t on the face F .

We will focus on these terms:

1.

$$D_t = \int_0^{t \wedge \tau_K} \mathcal{L}V(X_s) ds,$$

the interior drift term.

2.

$$M_t = \int_0^{t \wedge \tau_K} \nabla V(X_s)^\top \sigma(X_s) dW_s,$$

the diffusion term.

3.

$$K_t = \frac{1}{2} \sum_{i < j} \sum_{F \in \mathcal{F}_{ij}} \int_0^{t \wedge \tau_K} (\nabla V_j(X_s) - \nabla V_i(X_s)) \cdot n_{i \rightarrow j, F} dL_s^F,$$

the interface term.

We need to prove that

$$D_t + M_t + K_t$$

is a supermartingale.

Some natural assumptions, which are sufficient to guarantee that $V(X_{t \wedge \tau_K})$ is a supermartingale, are the following.

1. The interior generator is nonpositive on every cell:

$$\mathcal{L}V_i(x) \leq 0, \quad x \in \text{int}(K_i).$$

Equivalently,

$$(p_i + Q_i x) \cdot f(x) + \frac{1}{2} \text{tr}(a(x) Q_i) \leq 0, \quad x \in \text{int}(K_i).$$

This is the multidimensional analogue of the condition

$$f(x)V'(x) \leq 0$$

from the one-dimensional piecewise linear case.

Fact

If $\mathcal{L}V(X_s) \leq 0$ for all $s \leq \tau_K$, then the drift term D_t is a supermartingale.

Proof: We want to prove

$$\mathbb{E}[D_t \mid \mathcal{F}_r] \leq D_r, \quad r \leq t.$$

By definition,

$$\mathbb{E}[D_t \mid \mathcal{F}_r] = \mathbb{E} \left[\int_0^{t \wedge \tau_K} \mathcal{L}V(X_s) ds \mid \mathcal{F}_r \right] \quad (52)$$

$$= \int_0^{r \wedge \tau_K} \mathcal{L}V(X_s) ds + \mathbb{E} \left[\int_{r \wedge \tau_K}^{t \wedge \tau_K} \mathcal{L}V(X_s) ds \mid \mathcal{F}_r \right]. \quad (53)$$

Since $\mathcal{L}V(X_s) \leq 0$, the second term is nonpositive. Hence

$$\mathbb{E}[D_t \mid \mathcal{F}_r] \leq D_r.$$

Therefore D_t is a supermartingale.

2. The stochastic integral M_t is a martingale. A sufficient condition is

$$\mathbb{E} \left[\int_0^{t \wedge \tau_K} \|\nabla V(X_s)^\top \sigma(X_s)\|^2 ds \right] < \infty.$$

Since V is piecewise quadratic on finitely many cells, ∇V is piecewise linear. Therefore, on a compact set K , ∇V is bounded. Thus this condition is usually easy to verify if σ is bounded on K .

3. The interface contribution is nonpositive. The Tanaka interface condition is imposed face by face: for every $F \in \mathcal{F}_{ij}$, we assume

$$(\nabla V_j(x) - \nabla V_i(x)) \cdot n_{i \rightarrow j, F} \leq 0, \quad x \in F.$$

Since

$$\nabla V_i(x) = p_i + Q_i x$$

and

$$\nabla V_j(x) = p_j + Q_j x,$$

this condition can be written explicitly as

$$(p_j - p_i + (Q_j - Q_i)x) \cdot n_{i \rightarrow j, F} \leq 0, \quad x \in F, \quad F \in \mathcal{F}_{ij}.$$

This is the multidimensional version of the concavity condition

$$V'_+(x_i) - V'_-(x_i) \leq 0$$

from the one-dimensional case.

Fact

If

$$(\nabla V_j(x) - \nabla V_i(x)) \cdot n_{i \rightarrow j, F} \leq 0 \quad \text{on } F,$$

for every $F \in \mathcal{F}_{ij}$, and L_t^F is nondecreasing in t , then the interface term K_t is a supermartingale.

Proof: We want to prove

$$\mathbb{E}[K_t \mid \mathcal{F}_r] \leq K_r, \quad r \leq t.$$

The interface term is

$$K_t = \frac{1}{2} \sum_{i < j} \sum_{F \in \mathcal{F}_{ij}} \int_0^{t \wedge \tau_K} \gamma_{ij, F}(X_s) dL_s^F,$$

where

$$\gamma_{ij, F}(x) = (\nabla V_j(x) - \nabla V_i(x)) \cdot n_{i \rightarrow j, F}.$$

By assumption,

$$\gamma_{ij, F}(x) \leq 0 \quad x \in F, \quad F \in \mathcal{F}_{ij}.$$

Also, dL_s^F is a nonnegative measure, because surface local time is nondecreasing. Therefore

$$\int_r^t \gamma_{ij, F}(X_s) dL_s^F \leq 0.$$

Hence the future increment of K_t after time r is nonpositive, and so

$$\mathbb{E}[K_t \mid \mathcal{F}_r] \leq K_r.$$

Therefore K_t is a supermartingale.

5.1.1 Proof of the supermartingale property

We now combine the three conditions.

By the multidimensional Itô–Tanaka formula,

$$V(X_{t \wedge \tau_K}) = V(X_0) + D_t + M_t + K_t. \tag{54}$$

The drift term D_t is a supermartingale by the interior generator condition:

$$\mathcal{L}V_i(x) \leq 0 \quad x \in \text{int}(K_i).$$

The stochastic integral M_t is a martingale by the square-integrability assumption:

$$\mathbb{E} \left[\int_0^{t \wedge \tau_K} |\nabla V(X_s)^\top \sigma(X_s)|^2 ds \right] < \infty.$$

The interface term K_t is a supermartingale by the multidimensional concavity condition:

$$(\nabla V_j(x) - \nabla V_i(x)) \cdot n_{i \rightarrow j, F} \leq 0, \quad x \in F, \quad F \in \mathcal{F}_{ij}.$$

Therefore $D_t + K_t$ is a supermartingale, and M_t is a martingale. Since the sum of a martingale and a supermartingale is a supermartingale, we conclude that

$$V(X_{t \wedge \tau_K})$$

is a supermartingale.

5.1.2 Interpretation of the multidimensional concavity condition

The condition

$$(\nabla V_j(x) - \nabla V_i(x)) \cdot n_{i \rightarrow j, F} \leq 0, \quad x \in F, \quad F \in \mathcal{F}_{ij},$$

says that when the process crosses from cell K_i into cell K_j , the directional derivative of V in the crossing direction cannot increase.

Indeed, the normal derivative from the K_i side is

$$\nabla V_i(x) \cdot n_{i \rightarrow j, F},$$

while the normal derivative from the K_j side is

$$\nabla V_j(x) \cdot n_{i \rightarrow j, F}.$$

Thus the condition is equivalent to

$$\nabla V_j(x) \cdot n_{i \rightarrow j, F} \leq \nabla V_i(x) \cdot n_{i \rightarrow j, F}.$$

This is exactly the analogue of the one-dimensional condition

$$V'_+(x_i) \leq V'_-(x_i).$$

So, in multiple dimensions, ordinary concavity is not the only thing we need to look at. What matters for the local time term is the jump in the normal derivative across each interface. If that jump is nonpositive, then the surface local time contribution pushes $V(X_t)$ downwards rather than upwards.

5.1.3 Summary of sufficient conditions

We have shown that the following assumptions are sufficient.

1. V is continuous and piecewise quadratic:

$$V(x) = c_i + p_i^\top x + \frac{1}{2} x^\top Q_i x, \quad x \in K_i.$$

2. The generator is nonpositive inside each cell:

$$(p_i + Q_i x) \cdot f(x) + \frac{1}{2} \text{tr}(a(x) Q_i) \leq 0, \quad x \in \text{int}(K_i).$$

3. The stochastic integral is a martingale:

$$\mathbb{E} \left[\int_0^{t \wedge \tau_K} \|\nabla V(X_s)^\top \sigma(X_s)\|^2 ds \right] < \infty.$$

4. The interface condition holds on every shared face:

$$(p_j - p_i + (Q_j - Q_i)x) \cdot n_{i \rightarrow j, F} \leq 0, \quad x \in F, \quad F \in \mathcal{F}_{ij}.$$

Under these assumptions,

$$V(X_{t \wedge \tau_K})$$

is a supermartingale.

This gives the multidimensional version of the one-dimensional piecewise linear certificate result. The interior generator condition controls the ordinary Itô drift, while the interface concavity condition controls the Tanaka local time terms.

5.2 Reach-avoid stopping convention

The exit-time result above is a general auxiliary result. For the reach-avoid verifier, we stop earlier. Let $X_\tau \subseteq K$ be the target and let β be the upper certificate level. The stopping time used by the verifier is

$$\tau = \inf \{t \geq 0 : X_t \in X_\tau \text{ or } X_t \in \partial K \text{ or } V(X_t) \geq \beta\}.$$

Thus touching the target, the boundary of the verification domain, or the super- β set is the stopping moment. In particular, the unsafe set is not a separate stopping event: the condition $V \geq \beta$ on the unsafe set ensures that a continuous path reaching it has already reached the super- β stopping set.

With this convention, the interior generator condition is needed only on

$$\{x \in \text{int}(K) \setminus X_\tau : V(x) < \beta\}.$$

For verification it is conservative to check its closure, replacing $V < \beta$ by $V \leq \beta$. Likewise, the normal-derivative jump condition is needed only on the portions of internal faces outside the target and in this sub- β basin. No generator or interface condition is required after the process has reached any of the three stopping sets.

The stopping convention also requires the domain-boundary separation condition

$$\inf_{x \in \partial K \setminus X_\tau} V(x) \geq \beta.$$

Together with

$$\sup_{x \in X_0} V(x) \leq \alpha, \quad \inf_{x \in X_u} V(x) \geq \beta, \quad \alpha < \beta,$$

these are the conditions implemented by the multidimensional piecewise-quadratic verifier. Applying the preceding Itô-Tanaka argument up to τ shows that $V(X_{t \wedge \tau})$ is a supermartingale.

5.3 Asymptotic completeness of piecewise quadratic certificates

The theorem below retains the stronger exit-time formulation: it assumes the generator inequality throughout K and therefore proves a supermartingale up to τ_K . To use its approximants as reach-avoid certificates under the convention above, one additionally needs a strict boundary margin for the smooth certificate on $\partial K \setminus X_\tau$; uniform approximation then preserves $V_h \geq \beta$ there.

Theorem (asymptotic completeness under a strict smooth certificate). Let $K \subset \mathbb{R}^2$ be compact and polygonal, and consider the stopped SDE

$$dX_t = f(X_t) dt + \sigma(X_t) dW_t, \quad a(x) = \sigma(x)\sigma(x)^\top,$$

stopped at

$$\tau_K = \inf\{t \geq 0 : X_t \notin K\}.$$

Assume that f and a are bounded on K . Suppose there exists a strict smooth certificate $U \in C^3(K)$ such that

$$\mathcal{L}U(x) = \nabla U(x) \cdot f(x) + \frac{1}{2} \text{tr}(a(x) \nabla^2 U(x)) \leq -\varepsilon \quad x \in K,$$

for some $\varepsilon > 0$. Assume also that the boundary/separation conditions hold with margin: for some $\delta > 0$,

$$\sup_{x \in X_0} U(x) \leq \alpha - \delta, \quad \inf_{x \in X_u} U(x) \geq \beta + \delta, \quad \alpha < \beta.$$

Let $\{\mathcal{T}_h\}_{h \downarrow 0}$ be a shape-regular sequence of refined triangulations of K supporting C^1 piecewise-quadratic spline interpolants $I_h U$ satisfying

$$\|I_h U - U\|_{C^2(K)} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Then, for all sufficiently small h , the function

$$V_h := I_h U$$

is a continuous C^1 piecewise-quadratic certificate satisfying

$$\mathcal{L}V_h(x) \leq -\frac{\varepsilon}{2} \quad x \in K,$$

and

$$\sup_{x \in X_0} V_h(x) \leq \alpha, \quad \inf_{x \in X_u} V_h(x) \geq \beta.$$

Consequently,

$$V_h(X_{t \wedge \tau_K})$$

is a supermartingale.

Proof. Since f and a are bounded on K , define

$$F := \sup_{x \in K} \|f(x)\|, \quad A := \sup_{x \in K} \|a(x)\|.$$

Let

$$E_h := V_h - U.$$

Then

$$\mathcal{L}V_h - \mathcal{L}U = \nabla E_h \cdot f + \frac{1}{2} \text{tr}(a \nabla^2 E_h).$$

Hence, for a constant $C_K > 0$ depending only on the dimension and the chosen matrix norm,

$$|\mathcal{L}V_h(x) - \mathcal{L}U(x)| \leq F \|\nabla E_h\|_{L^\infty(K)} + C_K A \|\nabla^2 E_h\|_{L^\infty(K)}.$$

Because

$$\|E_h\|_{C^2(K)} \rightarrow 0,$$

there exists $h_0 > 0$ such that, for all $h < h_0$,

$$|\mathcal{L}V_h(x) - \mathcal{L}U(x)| \leq \frac{\varepsilon}{2} \quad x \in K.$$

Since $\mathcal{L}U \leq -\varepsilon$, it follows that

$$\mathcal{L}V_h \leq -\varepsilon + \frac{\varepsilon}{2} = -\frac{\varepsilon}{2}.$$

Similarly, since $V_h \rightarrow U$ uniformly, for all sufficiently small h ,

$$\|V_h - U\|_{L^\infty(K)} \leq \delta.$$

Therefore

$$\sup_{X_0} V_h \leq \sup_{X_0} U + \delta \leq \alpha,$$

and

$$\inf_{X_u} V_h \geq \inf_{X_u} U - \delta \geq \beta.$$

It remains to check the nonsmooth interfaces. The spline V_h is C^1 across every internal face of the triangulation. Therefore, if two quadratic pieces V_i, V_j meet across a face F_{ij} , then

$$\nabla V_i(x) = \nabla V_j(x) \quad x \in F_{ij}.$$

Hence the Tanaka normal-derivative jump is

$$[\nabla V_j(x) - \nabla V_i(x)] \cdot n_{i \rightarrow j} = 0,$$

so every local-time contribution is zero.

Applying the multidimensional Itô–Tanaka formula gives

$$V_h(X_{t \wedge \tau_K}) = V_h(X_0) + \int_0^{t \wedge \tau_K} \mathcal{L}V_h(X_s) ds + \int_0^{t \wedge \tau_K} \nabla V_h(X_s) \sigma(X_s) dW_s.$$

The drift term is nonpositive because

$$\mathcal{L}V_h \leq -\frac{\varepsilon}{2} < 0.$$

The stochastic integral is a true martingale after stopping, since K is compact and V_h, σ are bounded on K . Therefore

$$V_h(X_{t \wedge \tau_K})$$

is a martingale plus a nonincreasing finite-variation process, and is hence a supermartingale. $\square$